

PYTHON VARIABLES AND RULES

Variables • Declaration • Initialization • Assignment • Naming Rules

Variables made simple.

Learn • Practice • Remember

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Define Variable?

PYTHON BASICS / 01

A variable is a name used to store a value.

Think of it like a labeled box.

```
name = "Bala"
```

name → variable | "Bala" → value

Variable Declaration in Python

PYTHON BASICS / 02

Python does not require a separate variable declaration.

We normally create a variable by assigning a value.

```
name = "Bala"
```

```
age = 22
```

Python automatically understands the value's type.

Initialization:

PYTHON BASICS / 03

Initialization means giving a value to a variable for the first time.

Example:

```
age = 22
```

```
age → variable    |    22 → initial value
```

Multiple Variable Assignment

PYTHON BASICS / 10

Python allows us to assign values to multiple variables in a single line.

```
name, age, course = "Bala", 22, "Python"
```

This is the same as:

```
name = "Bala"  
age = 22  
course = "Python"
```

Another Example:

```
x, y, z = 10, 20, 30
```

Output:

```
print(x)  
print(y)  
print(z)
```

Output:

```
10  
20  
30
```

Key Points:

- Multiple variables can be assigned at the same time.
- Each variable gets the corresponding value.
- The number of variables and values should match.

- ✓ a, b, c = 10, 20, 30 # Correct
- ✗ a, b = 10, 20, 30 # Error (values don't match)

Assignment Operator =

PYTHON BASICS / 04

The symbol `=` is called the assignment operator.

It means: store the value on the right in the variable on the left.

```
age = 22
```

Think: $\text{age} \leftarrow 22$

Changing a Variable

PYTHON BASICS / 05

A variable's value can be changed later.

```
age = 22
```

```
age = 23
```

After the second assignment, age contains 23.

Variable Naming Rules

PYTHON BASICS / 06

- ✓ **Can contain letters: name**
- ✓ **Can contain numbers: student1**
- ✓ **Can contain underscore: student_name**
- X Cannot start with a number: 1student
- X Spaces are not allowed: student name
- X Avoid Python keywords such as if, for, class, return

Python is Case-Sensitive

PYTHON BASICS / 07

name, Name and NAME are different variables.

Example:

```
name = "Bala"
```

```
Name = "Arun"
```

```
print(name) → Bala
```

```
print(Name) → Arun
```

Variables Can Store Different Values

PYTHON BASICS / 08

String: name = "Bala"

Integer: age = 22

Float: height = 5.8

Boolean: is_student = True

Use `type(variable)` to check the data type.

How Python Automatically Understands Data Types

PYTHON BASICS / 06

Python automatically determines the data type based on the value assigned to a variable.

```
name = "Bala"      # String
age = 22           # Integer
height = 5.8       # Float
is_student = True  # Boolean
```

How does Python identify them?

```
"Bala" → String (str)
22     → Integer (int)
5.8    → Float (float)
True   → Boolean (bool)
```

Check the Data Type

```
print(type(name))
print(type(age))
```

Output:

```
<class 'str'>
<class 'int'>
```



Key Point:

Python does not require us to declare the data type separately. It identifies the type from the assigned value.

Remember:

Python is dynamically typed.

Simple Student Example

PYTHON BASICS / 10

```
student_name = "Bala"
```

```
roll_no = 101
```

```
age = 22
```

```
course = "Python"
```

```
duration = 12
```

```
print(student_name)
```

Key Terms to Remember

PYTHON BASICS / 11

Variable → a name used to store a value

Declaration → Python usually creates the variable through assignment

Initialization → giving the first value

Assignment → storing a value using =

Reassignment → changing the stored value

Quick Practice

PYTHON BASICS / 12

Create variables for:

- Your name
- Your age
- Your course
- Your roll number

Then print all four variables.

Summary

PYTHON BASICS / 13

Remember:

1. Variable = name + value
2. Python does not need separate type declaration
3. Initialization = first value
4. = is the assignment operator
5. Follow variable naming rules